$$FUE(L) = \frac{\text{total \# undetectable error bursts of length } L}{\text{total \# diff error bursts of length } L}$$

$$= \frac{\text{total \# undetectable } L}{2^{L-2} * \underbrace{(n-L+1)}} \quad \begin{array}{l} \Rightarrow \text{depends on error detection code} \\ \Rightarrow \text{all error detection codes} \end{array}$$

E.g. single even parity check code

$k=4$, $n-k=1$, $n=5$

$$FUE(L=4) = \frac{4}{2^2 * 2} = \frac{4}{8} = 50\%$$

$e = [\_ \_ \_ \_ \_]_{1\times 5}$

$$\left.\begin{array}{l} 0\ 1\ ?\ ?\ 1 \Rightarrow \binom{2}{0} + \binom{2}{2} = \underbrace{2} \\ 1\ ?\ ?\ 1\ 0 \Rightarrow \binom{2}{0} + \binom{2}{2} = 2 \end{array}\right\} \underbrace{2+2=4}$$

① # starting positions : $n-L+1 = 2$

② # "?" terms $= L-2$ $\Rightarrow$ to produce even # bit errors, we must have even # "?" positions set to 1

$$\binom{L-2}{0} + \binom{L-2}{2} + \binom{L-2}{4} + \cdots.$$

$$\Rightarrow \left[\binom{L-2}{0} + \binom{L-2}{2} + \cdots\right] * (n-L+1) = \left[\binom{2}{0} + \binom{2}{2}\right] * 2 = 4$$

$$FUE(L=5) = \frac{\left[\binom{3}{0} + \binom{3}{2}\right] * (5-5+1)}{2^{5-2} * (5-5+1)}$$

$$= \frac{(1+3) * 1}{8 * 1}$$

$$= 50\%$$

CRC (Cyclic Redundancy Check) code

Polynomial code

" Binary Polynomial "

info bits $\rightarrow$ $\boxed{\text{encoder}}$ $\rightarrow$ codeword (B) "pattern" $\rightarrow$ channel (error (e) $\downarrow$) $\rightarrow$ received bits ($R = B + e$, + : XOR) $\rightarrow$ $\boxed{\text{decoder}}$ $\rightarrow$

* 6 steps at encoder

① Given k info bits $i_{k-1}, i_{k-2}, \cdots, i_1, i_0$ $\Rightarrow$ info polynomial $i(x)$

$$i(x) = i_{k-1} * x^{k-1} + i_{k-2} * x^{k-2} + \cdots + i_1 * x + i_0 * \underbrace{x^0}_{1}$$

$i(x)$ is a binary polynomial with a degree of up to $(k-1)$

② to generate $(n-k)$ check bits

$\Rightarrow$ pick a generator polynomial $g(x)$: known a priori to both sender and receiver

* $g(x)$: degree $(n-k)$

$$g(x) = \underbrace{g_{n-k}}_{\Downarrow\; g_{n-k} = 1} \cdot x^{n-k} + g_{n-k-1} \cdot x^{n-k-1} + \cdots + g_1 \cdot x^1 + g_0 \cdot \underbrace{x^0}_{1}$$

* $g_0 = 1$

E.g. $n-k = 3$

$g(x) = x^3 + 1$ (valid)

$g(x) = x^3 + x^2 + 1$ (valid)

$g(x) = x^3 + x$ (invalid)

③ Long Division

dividend polynomial $a(x) = i(x) * x^{n-k}$

④ divisor polynomial : $g(x)$

to find out the remainder polynomial : $\underbrace{r(x)}$

$$a(x) = g(x) * \underbrace{q(x)}_{\text{quotient polynomial}} + \underbrace{r(x)}_{\deg(r(x)) < \deg(g(x)) = n-k}$$

⑤ codeword polynomial

$$B(x) = \underbrace{a(x)} + r(x) = i(x) \cdot x^{n-k} + r(x)$$

⑥ $B(x)$ coefficients => $\underbrace{n}$ codeword bits

All coefficient calculations follow "binary polynomial arithmetic"

$\underbrace{\text{2's complement arithmetic}}$

1 + 1 = 0	0 + 1 = 1	1 + 0 = 1	0 + 0 = 0
1 - 1 = 0	$\underbrace{0 - 1 = 1}$	1 - 0 = 1	0 - 0 = 0

E.g. $(x^3 + x^2 + 1) - (x^4 + x^3 + x + 1)$

$= x^4 + x^2 + x$

$(x^2 + 1) + (x^2 + 1)$

$= 0$

$f(x) + f(x)$ regardless of $f(x)$

$= 0$

$i_3 \; i_2 \; i_1 \; i_0$

Example: $\quad k = 4 \quad : \quad (1\,1\,0\,1)$

$n - k = 3 \quad : \quad$ pick $g(x) = x^3 + x + 1$

$n = 7$

① $i(x) = i_3 \cdot x^3 + i_2 + x^2 + i_1 \cdot x + i_0$

$= x^3 + x^2 + 1$

② $g(x) = x^3 + x + 1$ as an example

③ $a(x) = i(x) \cdot x^{n-k} = (x^3 + x^2 + 1) \cdot x^3 = x^6 + x^5 + x^3$

④ divisor polynomial : $g(x) = x^3 + x + 1$

Long Division :

$x^3 + x^2 + x + 1 \leftarrow q(x)$

$g(x) \rightarrow x^3 + x + 1 \;\big)\; x^6 + x^5 \quad + x^3 \leftarrow a(x)$

$x^6 \quad + x^4 + x^3$

$x^5 + x^4$

$x^5 \quad + x^3 + x^2$

$x^4 + x^3 + x^2$

$x^4 \quad + x^2 + x$

$x^3 \quad + x$

$x^3 \quad + x + 1$

$1 = r(x)$

$a(x) = g(x) \cdot q(x) + r(x)$

(5) $B(x) = a(x) + r(x)$

$= \underbrace{x^6 + x^5 + x^3}\ + \underbrace{1}$

(6) $B(x) = 1\cdot x^6 + 1\cdot x^5 + 0\cdot x^4 + 1\cdot x^3 + 0\cdot x^2 + 0\cdot x + 1$

$\Rightarrow \quad B = [\,1 \;\; \underbrace{1 \;\; 0 \;\; 1}_{\text{info bits}} \;\; \underbrace{0 \;\; 0 \;\; 1}_{\text{check bits}}\,]_{1\times 7}$